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Studierichting: Informatica
Oefeningenreeks 7A nr. 11.5

$$f : \Omega \subseteq \mathbb{R}^4 \to \mathbb{R}^2 : (w, x, y, z) \to \left(\cot \frac{w \ln|z|}{xy^2}, \ln\left|\frac{x \cos y}{(z-1)^2}\right|\right)$$

Voor $w$:

$$\frac{\partial f_1}{\partial w} = -\csc^2\left(\frac{w \ln|z|}{xy^2}\right) \cdot \frac{\ln|z|}{xy^2}$$
$$\frac{\partial f_2}{\partial w} = \frac{(z-1)^2}{x \cos y} \cdot 0 = 0$$

Voor $x$:

$$\frac{\partial f_1}{\partial x} = \csc^2\left(\frac{w \ln|z|}{xy^2}\right) \cdot \frac{w \ln|z|}{x^2y^2}$$
$$\frac{\partial f_2}{\partial x} = \frac{(z-1)^2}{x \cos y} \cdot D_x\left(\frac{x \cos y}{(z-1)^2}\right) = \frac{(z-1)^2}{x \cos y} \cdot \frac{\cos y}{(z-1)^2} = \frac{1}{x}$$

Voor $y$:

$$\frac{\partial f_1}{\partial y} = -\csc^2\left(\frac{w \ln|z|}{xy^2}\right) \cdot D_y\left(\frac{w \ln|z|}{xy^2}\right) = \csc^2\left(\frac{w \ln|z|}{xy^2}\right) \cdot \frac{2w \ln|z|}{xy^3}$$
$$\frac{\partial f_2}{\partial y} = \frac{(z-1)^2}{x \cos y} \cdot \frac{-x \sin y}{(z-1)^2} = -\frac{\sin y}{\cos y} = -\tan y$$

Voor $z$:

$$\frac{\partial f_1}{\partial z} = -\csc^2\left(\frac{w \ln|z|}{xy^2}\right) \cdot \frac{w}{xy^2z}$$
$$\frac{\partial f_2}{\partial z} = \frac{(z-1)^2}{x \cos y} \cdot \left(-\frac{2x \cos y}{(z-1)^3}\right) = -\frac{2}{z-1}$$

Matrix met $u = \dfrac{w \ln|z|}{xy^2}$

$$Df(w, x, y, z) = \begin{pmatrix} -\csc^2(u) \cdot \frac{\ln|z|}{xy^2} & \csc^2(u) \cdot \frac{w \ln|z|}{x^2y^2} & \csc^2(u) \cdot \frac{2w \ln|z|}{xy^3} & -\csc^2(u) \cdot \frac{w}{xy^2z} \\ 0 & \frac{1}{x} & -\tan y & -\frac{2}{z-1} \end{pmatrix}$$

# References